Tan Kuok Xuen

Senior AI Engineer

Enterprise Generative AI • NLP • Applied Machine Learning

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Summary

Senior AI Engineer with 8+ years of software engineering experience across generative AI, NLP, cloud-native systems and production ML applications. Currently building enterprise AI workflows at Microsoft using Azure OpenAI, Azure AI Foundry, Cosmos DB, .NET and React. Previously delivered multilingual NLP systems, semantic search and recommendation platforms at Intel, with additional experience building applied LLM products, backend services and data platforms in startup environments.

Selected Highlights

- Building enterprise GenAI applications at Microsoft for customer intelligence, brief generation and incident investigation using Azure OpenAI and Azure AI Foundry.
- Improved Intel enterprise search relevance by 20% using transformer-based semantic search and ranking techniques.
- Completed a Master of Philosophy in Artificial Intelligence with research focused on NLP, knowledge graphs and support systems.
- Presented NLP work at AI Everywhere 2023 and Software Professionals Conference 2022.

Experience

Microsoft **Jan 2026 – Present**
Senior AI Engineer (Contract) *Remote*

- Build enterprise AI applications used for customer intelligence, brief generation and incident investigation using Azure OpenAI and Azure AI Foundry.
- Develop multi-step AI workflows that combine LLM reasoning, enterprise data retrieval and structured business logic for internal decision-support use cases.
- Build production React dashboards for health monitoring, subscriptions and intake workflows, replacing rule-based summaries with AI-generated operational insights.
- Design bulk migration and onboarding pipelines to backfill enterprise customer records into Cosmos DB with dry-run, rollback and idempotency safeguards.
- Integrate Cosmos DB, Kusto and GraphQL data sources to support AI-powered analytics and operational workflows across a .NET 8, React 18 and Azure stack.

Navigator **Mar 2024 – Nov 2025**
Machine Learning Engineer *Remote – Malaysia*

- Developed production AI features and backend services using FastAPI, OpenAI APIs and data workflows for internal product development.
- Engineered scalable data pipelines for external data acquisition and transformation, reducing manual delivery cycles from multi-day processes to near real-time access.
- Managed AWS infrastructure across ECS, Lambda, RDS, S3 and Docker-based deployments while keeping operational costs under control.
- Implemented CI/CD workflows with GitHub Actions to improve deployment reliability and reduce release friction.

Intel Corporation **Mar 2021 – Mar 2024**
Machine Learning Engineer *Penang, Malaysia*

- Developed multilingual NLP recommendation systems supporting manufacturing and support operations across 4+ countries.
- Improved enterprise search relevance by 20% using transformer-based semantic search and ranking techniques.
- Trained transformer models for manufacturing NLP use cases, achieving over 90% accuracy on targeted classification tasks.
- Collaborated with Intel Labs on language model research that progressed into internal proof-of-concept deployments.
- Applied knowledge graphs, machine learning and NLP techniques to improve incident management and knowledge discovery workflows.

Experience Continued

Crosslight Capital **Apr 2018 – Mar 2021**
Quantitative Developer *Kuala Lumpur, Malaysia*

- Developed and backtested systematic trading strategies in C#, with a focus on research tooling, execution logic and performance analysis.
- Built analytics dashboards to support trading research, execution review and operational workflows.

Configura **Sep 2017 – Apr 2018**
Application Developer *Kuala Lumpur, Malaysia*

- Developed software features for configurable design and planning solutions, contributing across implementation, testing and release activities.

Education

Monash University **Mar 2021 – Jan 2023**
Master of Philosophy – Artificial Intelligence

- Thesis: Natural Language Processing in Support Systems: Applications of Knowledge Graph and Machine Learning.

Monash University **Oct 2014 – Jun 2017**
Bachelor of Computer Science

Personal Projects

BlooBloo AI Dive Logger | *AI-assisted web application*

- Built bloobloo.vercel.app, a personal AI-powered web application for capturing and organizing recreational diving records.

Additional Experience

PADI **Oct 2025 – Dec 2025**
Divemaster Intern *Pulau Tioman, Pahang, Malaysia*

- Completed professional diving training during a planned career break and gained operational experience in a safety-critical environment.

Selected Presentations

AI Everywhere 2023 | *Natural Language Processing for Incident and Knowledge Management Systems* **Mar 2023**

Software Professionals Conference 2022 | *NLP System for Intel Premier Support* **Nov 2022**

Certifications and Courses

- Natural Language Processing Nanodegree – Udacity
- Building Cloud Computing Solutions at Scale – Coursera

Technical Skills

Languages: Python, C#, SQL, TypeScript

AI and ML: Azure OpenAI, Azure AI Foundry, OpenAI APIs, PyTorch, Hugging Face, Transformers, MLflow, Semantic Search, Recommendation Systems, Knowledge Graphs

Cloud and Infrastructure: Azure, AWS, Docker, GitHub Actions, CI/CD

Frameworks and Platforms: FastAPI, React, .NET, Cosmos DB, Kusto, MongoDB, SQL Server, PostgreSQL, Plotly Dash